

WEEK11-PRACTICAL

Task1:

Develop a multithreaded counter application using POSIX threads (pthread_create, pthread_join). Demonstrate race conditions by allowing multiple threads to update a shared variable concurrently.

```
UW PICO 5.09 File: race_counter.c Modified
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <unistd.h>

#define NUM_THREADS 5
#define ITERATIONS 1000000

int counter = 0;

void *increment_counter(void *arg)
{
    int thread_id = *(int *)arg;

    for (int i = 0; i < ITERATIONS; i++)
    {
        counter++;
    }

    printf("Thread %d finished\n", thread_id);

    return NULL;
}

int main()
{
    pthread_t threads[NUM_THREADS];
    int thread_ids[NUM_THREADS];

    printf("Starting race-condition test...\n");

    for (int i = 0; i < NUM_THREADS; i++)
    {
        thread_ids[i] = i + 1;

        if (pthread_create(
            &threads[i],
            NULL,
            increment_counter,
            &thread_ids[i]) != 0)
        {
            perror("pthread_create");
            return 1;
        }
    }

    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_join(threads[i], NULL);
    }

    int expected = NUM_THREADS * ITERATIONS;

    printf("\nExpected counter value: %d\n", expected);
    printf("Actual counter value: %d\n", counter);

    if (counter == expected)
    {
        printf("Counter is correct.\n");
    }
}
```

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```
UW PIC0 5.09 File: race_counter.c Modified

for (int i = 0; i < NUM_THREADS; i++)
{
    thread_ids[i] = i + 1;
    if (pthread_create(
        &threads[i],
        NULL,
        increment_counter,
        &thread_ids[i]) != 0)
    {
        perror("pthread_create");
        return 1;
    }
}

for (int i = 0; i < NUM_THREADS; i++)
{
    pthread_join(threads[i], NULL);
}

int expected = NUM_THREADS * ITERATIONS;

printf("\nExpected counter value: %d\n", expected);
printf("Actual counter value: %d\n", counter);

if (counter == expected)
{
    printf("Counter is correct.\n");
}
else
{
    printf("Race condition detected!\n");
    printf("Some increments were lost.\n");
}

return 0;
}

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```

```
vishalkarnati@VISHALS-MacBook-Air os % mkdir week11pract
vishalkarnati@VISHALS-MacBook-Air os % cd week11pract
vishalkarnati@VISHALS-MacBook-Air week11pract % nano race_counter.c
vishalkarnati@VISHALS-MacBook-Air week11pract % gcc race_counter.c -o race_counter -pthread
vishalkarnati@VISHALS-MacBook-Air week11pract % ./race_counter
Starting race-condition test...
Thread 2 finished
Thread 1 finished
Thread 3 finished
Thread 4 finished
Thread 5 finished

Expected counter value: 5000000
Actual counter value: 1996536
Race condition detected!
Some increments were lost.
vishalkarnati@VISHALS-MacBook-Air week11pract %
```

Task2:

Modify the previous program using mutex locks (`pthread_mutex_lock`, `pthread_mutex_unlock`) and compare the outputs with the race-condition version. Analyze the impact of synchronization.

```
UW PICO 5.09 File: mutex_counter.c Modified
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>

#define NUM_THREADS 5
#define ITERATIONS 1000000

int counter = 0;
pthread_mutex_t lock;

void *increment_counter(void *arg)
{
    int thread_id = *(int *)arg;

    for (int i = 0; i < ITERATIONS; i++)
    {
        pthread_mutex_lock(&lock);
        counter++;
        pthread_mutex_unlock(&lock);

        printf("Thread %d finished\n", thread_id);
    }
    return NULL;
}

int main()
{
    pthread_t threads[NUM_THREADS];
    int thread_ids[NUM_THREADS];

    /*
     * Initialize mutex.
     */
    if (pthread_mutex_init(&lock, NULL) != 0)
    {
        perror("pthread_mutex_init");
        return 1;
    }

    printf("Starting mutex-protected counter test...\n");
    for (int i = 0; i < NUM_THREADS; i++)
    {
        thread_ids[i] = i + 1;

        if (pthread_create(
            &threads[i],
            NULL,
            increment_counter,
            &thread_ids[i]) != 0)
        {
            perror("pthread_create");
            return 1;
        }
    }
}
```

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```
UW PIC0 5.09 File: mutex_counter.c Modified
{
    pthread_t threads[NUM_THREADS];
    int thread_ids[NUM_THREADS];

    /*
     * Initialize mutex.
     */
    if (pthread_mutex_init(&lock, NULL) != 0)
    {
        perror("pthread_mutex_init");
        return 1;
    }

    printf("Starting mutex-protected counter test...\n");

    for (int i = 0; i < NUM_THREADS; i++)
    {
        thread_ids[i] = i + 1;

        if (pthread_create(
            &threads[i],
            NULL,
            increment_counter,
            &thread_ids[i]) != 0)
        {
            perror("pthread_create");
            return 1;
        }
    }

    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_join(threads[i], NULL);
    }

    int expected = NUM_THREADS * ITERATIONS;
    printf("\nExpected counter value: %d\n", expected);
    printf("Actual counter value: %d\n", counter);

    if (counter == expected)
    {
        printf("Counter is correct.\n");
        printf("Race condition prevented by mutex.\n");
    }
    else
    {
        printf("Unexpected result.\n");
    }

    /*
     * Destroy mutex.
     */
    pthread_mutex_destroy(&lock);

    return 0;
}

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```

```
vishalkarnati@VISHALS-MacBook-Air: week11pract % nano mutex_counter.c
vishalkarnati@VISHALS-MacBook-Air: week11pract % gcc mutex_counter.c -o mutex_counter -pthread
vishalkarnati@VISHALS-MacBook-Air: week11pract % ./mutex_counter
Starting mutex-protected counter test...
Thread 5 finished
Thread 2 finished
Thread 3 finished
Thread 1 finished
Thread 4 finished

Expected counter value: 5000000
Actual counter value: 5000000
Counter is correct.
Race condition prevented by mutex.
vishalkarnati@VISHALS-MacBook-Air: week11pract %
```

